



LAB ASSIGNMENT – 2

Performance Measurement Results

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Programme	BCA (AI & DS)	Semester	V
Course Code	ENCA351	Section	B
Course	Design and Analysis of Algorithms Lab	Faculty	Dr. Aarti

1. Record Execution Time, Memory Usage and Operation Counts

Measured benchmark values used in the lab. Values are taken from the generated benchmark data.

A. Search Algorithms

Algorithm	Input Size	Time (ms)	Peak Memory (KB)	Operations
Binary Search	100	0.0028	0.023	6
Linear Search	100	0.0090	0.164	100
Binary Search	1000	0.0052	0.094	9
Linear Search	1000	0.4668	0.207	1000
Binary Search	10000	0.0111	0.125	13
Linear Search	10000	6.7841	0.207	10000
Binary Search	50000	0.0169	0.125	15
Linear Search	50000	30.6135	0.207	50000

B. Loop Benchmark

Algorithm	Input Size	Time (ms)	Peak Memory (KB)	Operations
Single Loop Traversal	100	0.0300	0.102	100
Single Loop Traversal	500	0.2228	0.133	500
Single Loop Traversal	1000	0.4875	0.133	1000
Single Loop Traversal	2000	1.2387	0.133	2000
Single Loop Traversal	5000	2.9253	0.133	5000
Single Loop Traversal	10000	5.8907	0.133	10000
Single Loop Traversal	50000	37.6578	0.141	50000
Nested Loop Traversal	100	3.5340	0.156	10000
Nested Loop Traversal	500	127.5955	0.281	250000
Nested Loop Traversal	1000	581.9595	0.250	1000000



Algorithm	Input Size	Time (ms)	Peak Memory (KB)	Operations
Nested Loop Traversal	2000	2493.2772	0.281	4000000
Nested Loop Traversal	5000	16302.2818	1.531	25000000

C. Factorial Benchmark

Algorithm	Input n	Time (ms)	Peak Memory (KB)	Operations
Recursive Factorial	100	0.0656	0.195	100
Iterative Factorial	100	0.0351	0.234	100
Recursive Factorial	500	1.2217	8.090	500
Iterative Factorial	500	0.2547	1.105	500
Recursive Factorial	900	3.7962	20.590	900
Iterative Factorial	900	0.5464	2.090	900
Recursive Factorial	1000	4.9748	23.715	1000
Iterative Factorial	1000	0.6423	2.344	1000

Observation: Binary Search grows more slowly than Linear Search, a single loop scales more gently than a nested loop, and iterative factorial avoids recursion-stack growth.

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